

MANAV SHARMA

MERN STACK DEVELOPER · BACKEND & SYSTEM DESIGN

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EDUCATION

National Institute of Technology (NIT) Bhopal

B.Tech, Electronics & Communication Engineering | CGPA: 7.06/10

Coursework: DBMS, Operating Systems | Class XII (CBSE): 77.0% | Class X (CBSE): 92.0%

2023 – 2027

Bhopal, India

TECHNICAL SKILLS

Languages: C++, JavaScript (ES6+), Python

Backend: Node.js, Express.js, REST APIs, Socket.io, JWT, OAuth 2.0, Passport.js, RBAC, Bcrypt, Token Blacklisting, Razorpay

Data & Queues: MongoDB, Mongoose, PostgreSQL, MySQL, Redis, BullMQ

Infra & DevOps: Cloudflare R2, Cloudinary, PM2, GitHub Actions, CI/CD, Vercel, FFmpeg, HLS

Frontend: React.js, Redux Toolkit, Tailwind CSS, Three.js, GSAP, Framer Motion, Lenis

Testing & Tools: K6, Artillery, Autocannon, Clinic.js, Lighthouse, Commander.js, Archiver, Git, Postman, npm

PROJECTS

NexForge — Self-Hosted PaaS Platform

[Live](#) | [GitHub](#) | [npm](#)

Node.js, BullMQ, Redis, Socket.io, Express, React, GSAP + Lenis, GitHub OAuth, PM2, Razorpay

- **Architected a self-hosted PaaS backend** exposing build, deployment, and telemetry APIs with JWT-secured authentication, Archiver-based artifact compression, environment-variable sync, and zero-downtime rollback; shipped an npm CLI (1,076+ weekly organic downloads) for one-command application deployment.
- **Engineered a concurrent CI/CD pipeline** using BullMQ+Redis to orchestrate build jobs, with PM2 lifecycle management and symlink-based artifact swapping for zero-downtime deployments and instant rollbacks.
- **Automated the end-to-end deployment workflow** — from CLI archive upload and PAT authentication, through Redis/BullMQ build queues and worker-side npm builds with Cloudflare R2 artifact storage, to MongoDB route updates, an Express reverse proxy, and Socket.io real-time build-log telemetry streamed to both the terminal UI and web dashboard.
- **Monetized the platform with Razorpay subscriptions** — Pro tier enforcing plan-gated resource limits (deployments, custom domains) uniformly across the web dashboard and CLI, with webhook-verified payments and atomic free-tier quota checks.
- **Validated stability, security, and performance** — load-tested with Autocannon, Artillery, and K6, processing 1M+ requests at 2,802 req/s with 100% success rate (P99: 193ms) and zero server errors; profiled with Clinic.js across 42K+ samples confirming zero memory leaks and a stable event loop under peak load; hardened with a Redis-backed distributed rate limiter blocking traffic at the 50K request threshold; frontend delivered Lighthouse 99 Performance with sub-second Core Web Vitals (FCP: 0.5s, LCP: 0.8s, CLS: 0).

Adaptive Bitrate HLS Video Streaming Platform

[Live](#) | [GitHub](#)

React, GSAP + Lenis, Fluent-FFmpeg, BullMQ, Redis, Cloudflare R2, MongoDB, Razorpay

- **Built an adaptive-bitrate HLS transcoding engine** supporting 1080p/720p/480p playback with multi-audio tracks and soft subtitles — automating the full ingestion workflow from raw video upload, ffprobe media analysis, and multi-bitrate transcoding to R2 chunk storage and HLS master-playlist delivery, with live transcode progress streamed over Socket.io.
- **Monetized the platform with Razorpay-powered subscriptions** — tiered monthly and yearly plans gate playback to the plan's max resolution (480p/720p/1080p), premium content access, and download permissions via plan-aware rendition selection.
- **Designed a decoupled, scalable video pipeline** by offloading CPU-intensive transcoding to BullMQ worker nodes and storing HLS chunks on Cloudflare R2, keeping heavy workloads off the main API request path for improved responsiveness and reduced bandwidth costs.
- **Engineered an optimized FFmpeg split-filter pipeline** that decodes HEVC 10-bit video once and simultaneously encodes 1080p/720p/480p H.264 streams, saving 50%+ CPU over sequential encoding; fine-tuned presets and thread counts for 512MB RAM servers, reducing processing time by ~45% (4h → 2.25h) with zero OOM crashes, while auto-generating MP4 fallbacks, thumbnail sprite sheets, and R2 segment batches.
- **Load-tested with Autocannon, Artillery, and K6** across 1,000 concurrent VUs — processed 187K+ requests at a sustained 480 req/s with 100% success rate and zero dropped requests; frontend scored Lighthouse 90 Performance (FCP: 0.6s, CLS: 0) with jank-free GSAP + Lenis scrolling on high-refresh displays.

Crochella — Multi-Vendor 3D E-Commerce

[Live](#) | [GitHub](#)

React, Three.js, Node.js, Socket.io, BullMQ, Razorpay, Redis

- **Engineered a multi-tenant e-commerce marketplace** with 3 role-based dashboards (customer/admin/delivery), automating complex order splits, Razorpay webhook verification, automated PDF invoice generation, and delayed escrow payouts via a BullMQ-driven pipeline.
- **Architected a real-time delivery tracking system** using Socket.io and Leaflet maps for geospatial state synchronization, integrated with Cloudinary CDN and offloaded heavy workloads through BullMQ queues.
- **Developed an interactive 3D product catalog** using Three.js, GSAP ScrollTrigger, Lenis, and Framer Motion, optimizing GPU rasterization to maintain a flawless 120+ FPS during complex scroll animations on high-refresh displays.
- **Optimized backend throughput to 635 req/s** (50.9ms median, P99: 133ms) with a 100% success rate across 1,000 concurrent VUs, validated via Autocannon, Artillery, and K6 load testing.
- **Hardened system security and stability** with Redis rate-limiting (deflecting 3,000+ brute-force attempts), NoSQL injection prevention, and Clinic.js profiling confirming a flat ~50MB heap with zero memory leaks under sustained load; scored 100/100 on Lighthouse Best Practices.

ACHIEVEMENTS & LEADERSHIP

- **LeetCode:** Solved 500+ problems (287+ Medium, 82+ Hard) with 1,600+ Contest Rating; maintained 238-day streak demonstrating consistent problem-solving practice.
- **Robu Createo 3.0:** Secured 3rd Prize at national-level robotics and technical competition with 100+ participating engineering teams.
- **Technical Executive, Evolve Club NIT Bhopal** (March 2024 – Present): Led project development teams, coordinated technical workshops, mentored junior members; led team to 3rd Prize at Robu Createo 3.0.